

SCRAP & CHROME — Cyberpunk Systems Design

Design Philosophy

What started as a crafting outline has grown into a full systems doc: crafting and its Jank System, combat resolution, an economy, factions, base building, vehicles, a day-night cycle, and now player attributes and archetypes. The throughline hasn't changed — three tiers of scavenged material, five crafting stations, and the idea that anything can be built out of the "wrong" parts, but it'll show. Everything since has been built to plug into that same premise rather than sit beside it.

(Working title — rename freely if a better one comes to mind for the actual game.)

Table of Contents

1. [Material Tiers](#)
2. [Crafting Stations](#)
3. [The Jank System](#)
4. [Full Recipe List](#)
5. [Progression Hooks](#)
6. [Scavenge Locations](#)
7. [Random Encounters — Enemies](#)
8. [Weapon Stats](#)
9. [Armor Stats](#)
10. [Ammo Crafting](#)
11. [Carry Weight & Encumbrance](#)
12. [Craft Time](#)
13. [Skill Progression — Crafting Skill Trees](#)
14. [Economy — Barter & Coin Value](#)
15. [Commercial Locations](#)
16. [Repair Kits](#)
17. [Base Building](#)
18. [Vehicles](#)
19. [Factions & Reputation](#)
20. [Combat Resolution](#)
21. [Non-Crafting Skills](#)
22. [Cyberware & Humanity](#)
23. [Day-Night & Patrol Cycles](#)
24. [Player Attributes](#)
25. [Preset Character Types](#)

Material Tiers

Tier 1 – Junk (found everywhere, no tools needed)

Material	Common sources	Weight (kg)
Scrap metal	Cars, appliances, fences, dumpsters	1.0
Copper wire	Wall wiring, old electronics, motors	0.2
Plastic scrap	Bottles, casings, furniture	0.3
Cloth/fabric	Clothing, curtains, upholstery	0.2
Glass shards	Windows, bottles	0.3
Rubber	Tires, hoses, gaskets	0.5
Wood	Pallets, furniture, doors	1.5
Duct tape / adhesive	Toolboxes, garages	0.1
Chemical scraps	Cleaning supplies, batteries (acid), fertilizer	0.3
Bone/leather (organic)	Corpses, animal carcasses, tanneries	0.4
Spring/coil	Mattresses, pens, mechanisms	0.2
Razor blades	Bathrooms, box cutters, disposable razors	0.05
Razorwire	Security fences/perimeters, industrial sites	0.8 (per coil)
Lighter	Convenience stores, kitchens, corpses	0.1
Aerosol spray (flammable)	Bathrooms, garages, hardware stores	0.4
Nails	Toolboxes, construction sites, wood pallets	0.05 (per handful)
Fuel canister (gasoline)	Vehicles, garages, generators	2.0
Ball bearings / marbles	Toy stores, mechanical parts, bearings in machinery	0.2 (per bag)
Fishing line / cord	Sporting goods, docks, tackle boxes	0.1
Bleach / caustic chemicals	Bathrooms, cleaning supply closets, janitor rooms	0.5
Matches	Kitchens, convenience stores, camping gear	0.05
Gunpowder	Fireworks, hunting/ammo stores, homemade (saltpeter+charcoal+sulfur)	0.1 (per unit)
Used shell casing	Battlefields, shooting ranges, after firefights, corpses	0.02

Tools (reusable, not consumed by crafting)

Tool	Common sources	Used for	Weight (kg)
Wire cutters	Toolboxes, electrical rooms, garages	Razor whip, wiring/electronics prep	0.4
Soldering iron	Electronics stores, workbenches	Tech Rig recipes	0.6
Screwdriver/wrench set	Garages, workbenches	General Workbench recipes	1.0

Tier 2 – Refined / Intermediate (needs a workbench or basic tool to extract)

Material	Sources	Weight (kg)
Steel plating	Stripped vehicle panels, riot shields	3.0
Circuit boards	Any dead electronic device	0.3
Batteries (lead-acid, li-ion)	Cars, drones, phones	1.0
Small motors/servos	Fans, RC toys, power tools	0.8
Optics/lenses	Cameras, glasses, binoculars	0.3
Fiber optic cable	Corp infrastructure, dead terminals	0.2
Coolant/lubricant	Vehicles, HVAC systems	0.5
Explosive compounds	Mining sites, military caches, fertilizer + chem scrap	0.6
Reinforced polymer	Riot gear, drone shells	0.7
Basic chips/processors	Appliances, cheap consumer tech	0.1
Capacitors	TVs, power supplies, arcade machines	0.3
Primer (percussion cap)	Ammo crafting kits, gun stores, hardware stores	0.01

Tier 3 – High-Tech / Rare (needs a specialized bench + often a Tier 2 prerequisite)

Material	Sources	Weight (kg)
AI cores / neural chips	Corp security bots, high-end drones, black market	0.5
Cybernetic components	Dead cyborgs, ripperdoc clinics, corpse looting	1.0
Fusion/energy cells	Military hardware, corp power stations	1.5
Nanite paste	Medical facilities, corp labs	0.3
Smart-alloy (memory metal)	High-end vehicles, corp armor	1.2

Material	Sources	Weight (kg)
Quantum/encrypted chips	Corp servers, black ICE hardware	0.2
Biotech tissue samples	Cloning labs, bio-corp facilities	0.4
Graphene sheeting	Aerospace scrap, prototype tech	0.3
Holo-emitters	Advertising drones, AR gear	0.4

Crafting Stations

1. **Scrap Bench** (no power needed) — Tier 1 recipes: shivs, patch armor, basic tools, traps.
2. **Workbench** (needs power/tools) — Tier 1+2: firearms mods, melee weapons, basic armor, utility gadgets, ammo assembly.
3. **Tech Rig** (needs a soldering setup + processor) — Tier 2+3: electronics, drones, hacking tools, comms, fusion cell recharging.
4. **Chem Lab** — Tier 1+2 chemicals: meds, drugs, explosives, poisons.
5. **Ripperdoc Chair** (black-market or player-built) — Tier 3: cybernetics, bio-mods, implants.

Fusion Cell Recharging

Fusion/energy cells are **rechargeable**, not single-use consumables — plug a depleted cell into a Tech Rig with an active power source (base generator or solar rig) to top it back off. A recharge costs no materials, just time (see Craft Time below) and, if your base tracks power as a resource, a chunk of that power pool. Players without a powered base yet can still hot-swap a depleted cell for a fresh looted one instead of waiting.

The Jank System

Every recipe has an **ideal build** (the "real" ingredient list) and a set of **valid substitutes** one tier down. You can always build the item with substitutes — you just pay for it in reliability. This is the core tension of the scavenger fantasy: you *can* railgun someone with a car battery, it's just going to try to kill you too.

Jank Rating (0–100)

Every crafted item has a hidden Jank Rating, calculated at build time:

- **Base jank:** 0 for an ideal build using every specified material.
- **+15 to +30 per substituted ingredient**, depending on how far off-tier the substitute is (Tier 2 material subbed for Tier 3 = worse than Tier 2-for-Tier-2 swap).
- **+10 if built without a schematic** (freehanding it from observed/guessed recipes).
- **–10 to –25 if built at a higher-tier station than required** (a Tier 2 gadget built at the Tech Rig instead of the Workbench gets cleaner work).
- **–5 to –20 based on crafting skill/perk investment.**
- **–15 if a Stabilizer Component is included** (see below).

Jank Tiers & Effects

Jank Band	Label	Stat effect	Malfunction chance / use	Extra effects
0–20	Stable	100% of ideal stats	~0%	None
21–50	Rigged	85–95% of ideal stats	2–5%	Visible sparks/rattle; slightly louder/hotter
51–80	Janky	65–85% of ideal stats	8–15%	Degrades 2x faster; audible tell (whine, hiss, clank) that can alert enemies/NPCs
81–100	Deathtrap	40–65% of ideal stats	20–35%	Chance of <i>catastrophic failure</i> (see below); heavy noise/heat signature

Malfunction Outcomes (rolled per use, weighted by category)

- **Weapons:** jam/misfire (lose the turn), backfire (self-damage), reduced damage this shot.
- **Explosives:** fizzle (dud), early detonation (danger to the crafter), reduced blast radius.
- **Tech/gadgets:** glitch (function fails silently), short-out (temporary disable + minor shock), false signal (wrong data/alert).
- **Cybernetics:** tremor (accuracy penalty for a few seconds), HUD flicker (vision static), feedback jolt (minor HP loss).
- **Vehicle mods:** stall, overheat (temporary speed penalty), sensor dropout.

Catastrophic failure (only possible at Deathtrap tier, ~10% of malfunction rolls): the item is destroyed and deals damage/effect back at the user or nearby allies – a jank railgun that catastrophically fails might arc back through the welder, a jank pipe bomb might go off in your hand.

Reducing Jank

- **Schematics** – a found/bought blueprint for a specific recipe zeroes out the "no schematic" penalty and often unlocks a cleaner substitute path.
- **Stabilizer Components** – a craftable Tier 2 item (basically shock-mounts + heat sink + voltage regulator, built from rubber + coolant + a capacitor) that can be slotted into almost any recipe to knock off a flat jank chunk. Consumed on use.
- **Skill/Perks** – "Steady Hands," "Field Engineer," etc. reduce jank from substitutions specifically, representing the character getting good at fudging it.
- **Better station** – building a Rigged item at a nicer station than required cleans it up (real workbench vs. a curb).
- **Repair/re-tune** – spending downtime and a small material cost at a matching station lets you "tune" a janky item down one band, representing field maintenance.

Handling Risk – Self-Inflicted Injury (separate from Jank)

Some weapons are just dangerous to use *by design*, regardless of build quality – exposed blades, no guard, no proper grip. These carry a baseline **Handling Risk** %: a flat chance, rolled on use (usually on a miss or a "hard" attack), that the wielder takes a small self-inflicted injury (cut hand, dropped weapon, minor bleed status). This stacks additively with any Jank-derived malfunction chance rather than replacing it – a jank-built razor whip is dangerous *twice over*.

Weapon	Base Handling Risk	Why
Broken-bottle knife	15%	Jagged edge, no guard
Nail bat	10%	Exposed nails can catch on the backswing
Pipe shiv	5%	Small, low leverage, minor risk
Razorclub	15%	Blades bound to a stick, no real hilt
Razor whip	20%	Flexible razorwire is as dangerous to the user as the target
Monowire garrote	25%	Monofilament edge cuts anything it touches, including a slipped grip
Fishing-line garrote	15%	Thin cord bites into hands under tension
Molotov cocktail	10%	Premature ignition on throw, especially if jank
Bear trap	10%	Careless disarm/reset snaps the jaws early
Jury-rigged chainsaw	20%	Kickback from an unbalanced, improvised build
Choke-gas bottle	15%	Fumes back at the thrower in close quarters/enclosed spaces
Aerosol flamethrower	10% (on jank/malfunction only)	Backflash risk if the spray/ignition timing is off

A Stabilizer Component or a proper **grip/guard upgrade** (leather wrap, tape, salvaged handle — cheap Tier 1 add-on) can reduce Handling Risk by roughly a third to half; it's a good cheap early-game crafting upgrade path that doesn't require new materials, just better assembly.

Ammo/Fuel-Tracked Weapons

A few weapons don't fire "ammo" in the traditional sense — they consume a fuel or feedstock material at a set rate instead of needing per-shot bullets crafted. Worth tracking separately since running dry mid-fight (rather than just reloading) is part of their risk/reward:

Weapon	Consumable(s)	Approx. uses per unit
Aerosol flamethrower	Lighter (ignition) / Aerosol spray (fuel)	~100 / ~50
Molotov cocktail	Fuel canister + matches/lighter	1 per bottle (single-use throwable)
Improvised slingshot	Ball bearings/marbles/rocks	Effectively unlimited (scavenge anywhere)
Improvised crossbow	Bolts (craftable)	1 per bolt, bolts are recoverable ~50% of the time
Jury-rigged nail gun	Nails	1 per nail, cheap to restock

Weapon	Consumable(s)	Approx. uses per unit
Jury-rigged chainsaw	Fuel canister	~40 uses
Pepper spray	Aerosol can	~30 uses

Design Notes

- Jank should be **visible before combat**, not a surprise — show it in the crafting menu (a wrench/sparks icon scaled 1–4) and on the item tooltip, so players are choosing risk, not getting ambushed by it.
- High-tier gear built *without* substitutes should still feel best-in-slot — jank is a bridge for players who don't have the rare mats yet, not a permanent parallel economy.
- Consider letting NPCs/factions react to jank: a jank weapon might visibly smoke or spark, and some vendors/gangs could refuse to buy visibly janky gear, or offer less for it.
- Optional flavor: let jank items get **nicknamed** procedurally ("Sparky," "The Widowmaker," "Lucky Seven") once they survive N uses without catastrophic failure — turns a liability into a character.

Full Recipe List

Format: **Item** — Ideal recipe (*Station, Tier*) — [Jank substitute if built cheaper]

Weapons

- Pipe shiv** — scrap metal + cloth (*Scrap Bench, T1*) — Δ 5% self-injury
- Broken-bottle knife** — glass shards + cloth grip (*Scrap Bench, T1*) — Δ 15% self-injury
- Nail bat** — wood + scrap metal (nails) (*Scrap Bench, T1*) — Δ 10% self-injury
- Razorclub** — razorwire + wood, OR razor blades + wood (*Scrap Bench, T1*) — Δ 15% self-injury
- Razor whip** — razorwire + wire cutters (tool) (*Workbench, T1*) — Δ 20% self-injury
- Aerosol flamethrower** — lighter + aerosol spray (*Workbench, T1*) — fuel-limited: lighter (ignition source) good for ~100 uses, aerosol can (fuel) good for ~50 uses before empty; either can be swapped independently once depleted. High jank on rebuilds with a damaged nozzle/lighter (backflash risk, see Handling Risk)
- Molotov cocktail** — glass bottle + cloth (wick) + fuel canister + matches/lighter (*Scrap Bench, T1*) — thrown AoE fire; Δ 10% self-injury (premature ignition on throw)
- Fishing-line garrote** — fishing line/cord + handle grip (wood or cloth-wrapped) (*Scrap Bench, T1*) — cheap early-game strangle weapon; Δ 15% self-injury (thin cord bites into hands under tension)
- Improvised slingshot** — wood + rubber (*Scrap Bench, T1*) — ammo: ball bearings/marbles/rocks (near-infinite, scavenge almost anywhere)
- Improvised crossbow** — wood + scrap metal + spring (*Workbench, T1*) — ammo: bolts (craft cheaply from scrap metal + wood)
- Bear trap** — scrap metal + spring (*Workbench, T1*) — placed trap, high damage/immobilize; Δ 10% self-injury if disarmed/reset carelessly
- Zip gun** — pipe (scrap metal) + spring + tape (*Workbench, T1*)
- Stun baton** — pipe + copper wire + battery (*Workbench, T2*) — [sub: capacitor → weaker charge, higher jank]
- Sawn-off improvised shotgun** — scrap metal + steel plating + spring (*Workbench, T2*)
- Jury-rigged nail gun** — small motor + scrap metal + spring (*Workbench, T2*) — ammo: nails (cheap, common)

- **Jury-rigged chainsaw** — small motor + scrap metal + fuel canister (*Workbench, T2*) — ammo: fuel canister (~40 uses); high damage melee, very loud (alerts nearby enemies); Δ 20% self-injury (kickback)
- **Pepper spray** — aerosol spray + chemical scraps (*Chem Lab, T1*) — non-lethal disable/blind, ~30 uses per can
- **Smart-targeting scope** — optics + circuit board + basic chip (*Tech Rig, T2*) — [sub: basic chip → no aim-assist, cosmetic only]
- **EMP grenade** — battery + copper wire + capacitor + plastic casing (*Tech Rig, T2*)
- **Taser gauntlet** — copper wire + battery + reinforced polymer (*Tech Rig, T2*)
- **Railgun coil pistol** — smart-alloy + fusion cell + circuit board + steel plating (*Tech Rig, T3*) — [sub: fusion cell → car battery: high jank, overheats fast]
- **Monowire garrote** — graphene sheeting + smart-alloy wire + handle grip (*Tech Rig, T3*) — Δ 25% self-injury
- **Smart-tracking dart launcher** — AI core + small motor + basic chip + reinforced polymer (*Tech Rig, T3*)

Armor & Clothing

- **Scrap plate vest** — steel plating + leather/cloth + tape (*Scrap Bench, T1*)
- **Padded jacket (impact)** — cloth + rubber padding (*Scrap Bench, T1*)
- **Riot patchwork jacket** — reinforced polymer + cloth + basic chip (impact sensor) (*Workbench, T2*)
- **Reinforced boots** — rubber + steel plating (toe cap) (*Workbench, T1*)
- **Faraday-lined coat** (anti-EMP) — copper wire mesh + cloth + plastic scrap (*Workbench, T2*)
- **Thermoptic-lite cloak** — holo-emitters + fiber optic + graphene sheeting (*Tech Rig, T3*) — [sub: fiber optic → copper wire: cloak flickers/visibly glitches, high jank]
- **Subdermal plating** — smart-alloy + nanite paste + cybernetic components (*Ripperdoc, T3*)

Tech & Gadgets

- **Lockpick multitool** — scrap metal + wire (*Scrap Bench, T1*)
- **Crowbar** — scrap metal (*Scrap Bench, T1*)
- **Signal jammer** — circuit board + copper wire + battery (*Tech Rig, T2*)
- **Portable radio/comm** — circuit board + battery + basic chip (*Tech Rig, T2*)
- **Camera drone** — small motor + circuit board + optics + battery (*Tech Rig, T2*) — [sub: battery → lead-acid: heavier, shorter flight, loud motor]
- **Stabilizer Component** — rubber + coolant + capacitor (*Tech Rig, T2*) — used to reduce jank on other builds
- **Portable hacking deck** — basic chip + fiber optic + quantum chip (*Tech Rig, T3*)
- **AI-assisted drone brain** — AI core + quantum chip + fusion cell (*Tech Rig, T3*)
- **Holo-decoy projector** — holo-emitters + battery + basic chip (*Tech Rig, T3*)

Medical

- **Field bandage** — cloth + adhesive (*Scrap Bench, T1*)
- **Splint** — wood + cloth (*Scrap Bench, T1*)
- **Antiseptic salve** — chemical scraps + plant/organic matter (*Chem Lab, T1*)
- **Trauma stim** — chemical scraps + battery (electro-injector) + basic chip (*Chem Lab, T2*)
- **Nanite healing gel** — nanite paste + biotech tissue + chemical compound (*Chem Lab, T3*) — [sub: biotech tissue → organic scrap: weak, risk of infection debuff]

Consumables / Drugs

- **Homebrew stimulant** — chemical scraps + plant matter (*Chem Lab, T1*)

- **Synth-adrenaline shot** — chemical compound + battery-injector + basic chip (*Chem Lab, T2*)
- **Focus tabs (aim steadiness)** — chemical scraps + glass (casing) (*Chem Lab, T1*)
- **Neuro-booster (reflex drug)** — nanite paste + biotech tissue (*Chem Lab, T3*)

Explosives & Traps

- **Caltrops** — scrap metal (*Scrap Bench, T1*)
- **Tripwire alarm** — copper wire + scrap metal (cans) (*Scrap Bench, T1*)
- **Pipe bomb** — pipe (scrap metal) + explosive compound + wire (*Workbench, T2*) — [sub: explosive compound → fertilizer + chem scrap mix: high jank, dud chance]
- **Nail bomb** — pipe bomb components + nails (shrapnel) (*Workbench, T2*) — higher damage/wider spread than pipe bomb, same jank profile
- **Choke-gas bottle (bleach bomb)** — bleach + chemical scraps + glass casing (*Chem Lab, T2*) — thrown, area disable (choking/blindness debuff); ⚠ 15% self-injury if thrown at close range or used in an enclosed space
- **Proximity mine** — explosive compound + circuit board + basic chip (*Tech Rig, T2*)
- **Shaped charge (breaching)** — explosive compound + steel plating + reinforced polymer (*Workbench, T2*)
- **Smart mine (target-selective)** — proximity mine components + AI core (*Tech Rig, T3*)

Cybernetics & Bio-mods

- **Basic neural link (comms only)** — circuit board + fiber optic + copper wire (*Ripperdoc, T2*)
- **Reflex booster implant** — cybernetic components + nanite paste + basic chip (*Ripperdoc, T3*)
- **Optic HUD implant** — optics + AI core + fiber optic (*Ripperdoc, T3*)
- **Subdermal armor weave** — smart-alloy + graphene sheeting + biotech tissue (*Ripperdoc, T3*)
- **Grip-strength servo arm** — small motor + steel plating + cybernetic components (*Ripperdoc, T2/T3 hybrid*)

Vehicle / Drone Mods

- **Reinforced bumper** — scrap metal + steel plating (*Workbench, T1*)
- **Jury-rigged engine patch** — scrap metal + rubber + coolant (*Workbench, T1*)
- **Overclocked engine chip** — basic chip + coolant + quantum chip (*Tech Rig, T3*) — [sub: quantum chip → basic chip only: real overheat risk, classic jank case]
- **Autopilot module** — AI core + circuit board + small motor (*Tech Rig, T3*)
- **Nitro injector** — chemical compound + small motor + steel plating (*Workbench, T2*)

Base / Territory Defense

- **Barbed scrap fence** — scrap metal + wood (*Scrap Bench, T1*)
- **Motion alarm** — circuit board + battery + basic chip (*Tech Rig, T2*)
- **Auto-turret** — steel plating + small motor + AI core + fusion cell (*Tech Rig, T3*) — [sub: fusion cell → multiple batteries: short uptime, overheats, jank-fires on own team occasionally]
- **Barricade kit** — wood + scrap metal + tape (*Scrap Bench, T1*)

Progression Hooks

- Gate Tier 3 *ideal* recipes behind **finding a schematic** — the jank-substitute path is always open, so players without the schematic aren't blocked, just penalized.

- Tie rare materials to **factions/locations** — corp zones drop Tier 3 tech, gang turf drops Tier 1/2 in bulk, medical districts drop biotech — so scavenging routes double as world-building.
- Let a character's **playstyle identity** form around jank: a "gutter tech" build that leans into high-jank gear with perks that reduce catastrophic-failure odds, vs. a "corpo defector" build that only ever builds clean because they came in with schematics and good mats.

Scavenge Locations

Each location has a **loot profile**: base chance of finding *anything* per search, a tier-weight split (T1/T2/T3), and a handful of location-specific "signature" finds that show up more than the general table suggests. Risk rating is about what's likely to be guarding or roaming the place, not the loot itself.

Loot Table Legend

- **Find chance** — odds a given search action yields something at all.
- **Tier weights** — of the things you do find, how they split across T1/T2/T3.
- **Signature finds** — items/materials this location should be *known for*, at a boosted rate.
- **Risk** — Low / Medium / High / Very High (hostiles, structural danger, rads, etc.)

Location	Find chance	Tier weights (T1/T2/T3)	Signature finds	Risk
Junkyard	Very High (~85%)	70/28/2	Scrap metal, steel plating, rubber, small motors, vehicle parts	Low–Medium (scavenger gangs, guard dogs)
Abandoned factory	High (~70%)	45/45/10	Steel plating, industrial motors, chemical scraps, reinforced polymer, occasional explosive compound	Medium (structural collapse, squatters)
City dump	High (~75%)	65/30/5	Plastic scrap, cloth, glass, scrap metal, batteries, rare intact electronics	Medium (toxic pockets, territorial gangs)
Abandoned house	Medium (~55%)	55/35/10	Cloth, wood, basic chips (old appliances), batteries, personal meds	Low (occasional squatters)
Shack / squatter camp	Medium (~50%)	60/30/10	Cloth, wood, chemical scraps, homebrew drugs, jury-rigged tools (pre-jank items!)	Medium (defensive occupants)
Back alley	Low–Medium (~40%)	60/35/5	Glass shards, cloth, discarded weapons (low-tier), drug residue, occasional stash	Medium–High (muggers, dealers)
Library / records archive	Low (~30%) for materials,	40/40/20	Fiber optic (old server rooms), circuit boards, rare:	Low

Location	Find chance	Tier weights (T1/T2/T3)	Signature finds	Risk
	high for intel		schematics/blueprints as "finds" instead of materials	
Sewer / maintenance tunnels	Medium (~45%)	55/35/10	Copper wire, coolant, pipes, corp-runoff chemicals, occasional dumped contraband	Medium-High (feral wildlife, gangs using it as a route)
Parking garage	Medium (~45%)	50/40/10	Vehicle batteries, steel plating, rubber, small motors, abandoned vehicle electronics	Low-Medium
Corp office (mid-tier)	Medium (~50%), high value	20/50/30	Circuit boards, basic/quantum chips, optics, corp ID (quest item), capacitors	High (security drones, alarms)
Medical clinic / hospital (abandoned)	Medium (~50%)	25/45/30	Chemical compounds, nanite paste, biotech tissue, medical schematics, meds	Medium-High (contamination, security if still active)
Ripperdoc back-room (raided)	Low (~35%), high value	10/35/55	Cybernetic components, nanite paste, smart-alloy scraps, cyber-schematics	High (owner or rival gang defending it)
Electronics store (looted)	Medium (~55%)	30/55/15	Circuit boards, basic chips, batteries, optics, capacitors	Low-Medium
Drone/vehicle wreckage site	Medium (~50%), high value	20/40/40	AI cores, fusion cells, smart-alloy, graphene sheeting, small motors	High (still-active security response, fire)
Corp server farm / data center	Low (~30%), very high value	5/25/70	Quantum chips, AI cores, encrypted chips, holomitters	Very High (black ICE, security teams)
Military checkpoint (abandoned/raided)	Low (~35%), high value	15/35/50	Explosive compounds, fusion cells, reinforced polymer, weapon schematics	Very High (unexploded ordnance, patrols)
Gang den / clubhouse (raided)	Medium (~45%)	35/40/25	Weapons (pre-built, possibly janky), drugs, chemical scraps, cash/valuables	High (defenders, traps)
Chop shop	Medium-High (~60%)	30/55/15	Steel plating, small motors, vehicle batteries,	Medium (mechanics/gang defenders)

Location	Find chance	Tier weights (T1/T2/T3)	Signature finds	Risk
			coolant, reinforced polymer	
Landfill / toxic waste site	Medium (~50%)	50/35/15	Chemical scraps, plastic scrap, rare industrial chemicals, occasional biotech runoff	High (toxicity, mutated wildlife)

Design Notes on Locations

- **Signature finds should be discoverable, not guaranteed** — a library isn't a materials location, it's where you go *specifically* for schematics; frame it as a distinct search verb ("research" vs. "scavenge") if that fits your systems.
- **Risk should scale opposite to reliability, not just loot value** — the junkyard is safe *and* has good T1/T2 volume, which is why it's the default early-game loop; the payoff for corp/military/server locations is Tier 3 access, not raw volume.
- Consider a **"searched" state per location** (or per sub-area within one) that depletes loot chance temporarily and regenerates over in-game days, so players have a reason to rotate a route instead of camping one spot.
- **Ripperdoc back-rooms and chop shops** work well as small hand-placed "vendor-adjacent" locations — high value, low volume, often already fought-over by NPCs, so finding one first (or clearing it) feels like a mini-heist rather than a loot pinata.
- Tie a couple of locations to **quest-gated access** (military checkpoint, server farm) so Tier 3 materials aren't just a matter of grinding the junkyard long enough — they require the player to go somewhere dangerous and often story-relevant.

Random Encounters — Enemies

Baseline assumes a player at ~100 HP with starter gear, mid-single-digit damage per basic hit. Scale up/down to fit your actual combat numbers — the point is the *relative* spread between enemy types.

Stat Legend

- **HP** — health pool
- **Armor** — flat damage reduction per hit
- **Dmg** — typical hit damage, calculated from the enemy's assigned weapon
- **Spd** — Slow / Normal / Fast / Erratic
- **Threat** — 1 (trivial) to 5 (avoid without prep)
- **Drops** — ties back into the crafting/loot system

Weapon Loadouts & Proficiency

Human enemies fight with the **same weapon pool the player crafts from** — they're scavengers too. Each enemy type has a **Proficiency Multiplier** applied to their assigned weapon's base Damage/Accuracy (from the Weapon Stats tables) to reflect skill, not access — a Street kid swinging a pipe shiv doesn't hit as hard or as often as the player would with the same shiv. Bots and animals use inherent, non-craftable attack profiles instead (their hardware/bite isn't something you loot as a usable item, even if the bot itself drops materials).

Their **Loadout Jank** column uses the same bands as the player's Jank System – most scavenged human enemies carry gear that's Rigged or worse, since they're building from the same junk piles you are. This means a Raider's sawn-off can jam or backfire on *them* mid-fight, which is worth surfacing to the player (a visible spark/stumble) as a tell.

Enemy	Type	HP	Armor	Weapon (base Dmg/Acc)	Proficiency	Effective Dmg/Acc	Spd
Street kid	Human (opportunist)	25	0	Pipe shiv (10/70)	50%	5 / 35	Fast
Wirehead	Human (drug-addled)	40	0	Broken-bottle knife (12/65)	65% (±40% erratic swing)	5–13 / 25–55	Erratic
Burner	Human (unstable enforcer)	45	5 (scrap armor)	Molotov cocktail (25/65)	75%	19 / 49	Normal
Scavenger	Human (rival looter)	35	5	Zip gun (22/45)	65%	14 / 29	Normal
Raider	Human (organized gang)	60	10 (scrap plating)	Sawn-off shotgun (38/50)	85%	32 / 43	Normal
Minuteman	Human (local militia)	55	8	Improvised crossbow (24/68)	85%	20 / 58	Normal
Wild dog	Animal	20	0	Bite (innate, Pierce)	—	4–6 / 60	Fast
Rogue service bot	Bot (civilian-grade)	50	15 (light chassis)	Improvised tool-arm (innate, Blunt)	—	3–6 / 50	Slow

Enemy	Type	HP	Armor	Weapon (base Dmg/Acc)	Proficiency	Effective Dmg/Acc	Spd
Rogue security bot	Bot (corp security)	75	25 (plated)	Integrated stun-turret (innate, Shock/Ballistic)	—	14–20 / 75	Normal
Operative	Human (corp agent)	65	12 (armored coat)	Railgun coil pistol (70/80)	95%	67 / 76	Fast
Soldier	Human (military)	90	20 (hard armor)	Military rifle (55/85)	100%	55 / 85	Normal
Rogue kill bot	Bot (military-grade)	150	35 (heavy plating)	Integrated heavy weapon (innate, Ballistic/Burn AoE)	—	22–30 / 85	Slow–Normal

Encounter Notes

- **Threat should map to location risk** from the scavenge table: Junkyard/dump pull mostly from Street kid, Wild dog, Scavenger; Corp/server/checkpoint locations pull from Operative, Soldier, Rogue security/kill bot.
- **Bots as an EMP sink** — give EMP grenades/tasers bonus effect against all three bot types, so the Tier-2 EMP grenade recipe has a clear tactical reason to exist beyond raw damage.
- **Pack/squad enemies (Wild dog, Raider, Soldier) reward crowd-control crafts** — traps, proximity mines, and AoE explosives get their moment against these rather than single-target weapons.
- **Loot-table crossover:** tying enemy drops directly to the material list means killing a Rogue security bot at a corp office both fits the fiction and doubles as an alternate path to circuit boards/optics if the player would rather fight than search.
- Consider a **rare "named" variant** of Raider/Operative/Soldier per region — same stat block plus one signature ability/jank-free unique weapon — to make random encounters occasionally feel like a mini-event instead of pure filler.

Weapon Stats

Numbers below assume an **ideal (0 Jank) build**. Jankier versions of the same weapon keep these base numbers but apply the stat-reduction and malfunction-chance modifiers from the Jank System above, and

degrade (lose Durability) at the rates noted there. Range is straight-line effective range before serious accuracy drop-off. Damage/Accuracy are on a 1–100 scale; Durability is uses/hits before the item breaks or needs a repair pass.

Melee

Weapon	Damage	Accuracy	Range	Type	Durability	Weight (kg)
Pipe shiv	10	70	1m	Stab	40	0.5
Broken-bottle knife	12	65	1m	Slash	15	0.3
Nail bat	16	60	1.5m	Blunt	35	1.5
Razorclub	18	62	1.5m	Slash/Blunt	25	1.2
Fishing-line garrote	14	55	0.5m (grapple)	Slash/Strangle	10	0.1
Razor whip	20	58	3m	Slash	20	0.6
Stun baton	8 (+ shock stun)	75	1m	Shock	60 charges	1.0
Taser gauntlet	6 (+ stun)	80	1m (touch)	Shock	50 charges	0.8
Jury-rigged chainsaw	45	65	1.5m	Slash	30	4.5
Monowire garrote	55	70	0.5m	Slash	25	0.2

Ranged

Weapon	Damage	Accuracy	Range	Type	Durability	Weight (kg)
Improvised slingshot	9	55	12m	Blunt (kinetic)	60	0.4
Zip gun	22	45	8m	Ballistic	8	0.8
Improvised crossbow	24	68	20m	Pierce	45	2.0
Jury-rigged nail gun	20	62	12m	Pierce	55	2.5
Sawn-off shotgun	38 (close spread)	50	10m	Ballistic	30	3.0

Weapon	Damage	Accuracy	Range	Type	Durability	Weight (kg)
Smart-tracking dart launcher	30 (+ DoT)	90	35m	Pierce	40	2.2
Railgun coil pistol	70	80	40m	Kinetic/Ballistic	20	2.8
Military rifle (<i>loot-only, not craftable</i>)	55	85	60m	Ballistic	80	3.5

Thrown, Placed & Explosive

Weapon	Damage	Accuracy	Range	Type	Durability	Weight (kg)
Choke-gas bottle	5 (+ disable DoT)	60	8m	Toxic	1 (single-use)	0.5
Molotov cocktail	25 (+ burn DoT)	65	15m	Burn	1 (single-use)	0.8
EMP grenade	15 (organic) / severe vs. bots	85	12m	EMP	1 (single-use)	0.4
Pipe bomb	50 (AoE)	90 (placement)	10m (throw)	Explosive	1 (single-use)	0.6
Nail bomb	60 (AoE + shrapnel)	90 (placement)	10m (throw)	Explosive	1 (single-use)	0.7
Proximity mine	55	100 (trigger-based)	0 (3m trigger radius)	Explosive	1 (single-use)	1.0
Smart mine	65	100 (trigger, target-selective)	0 (5m trigger radius)	Explosive	1 (single-use)	1.2
Shaped charge	90 (vs. structures/armor)	100 (placement)	0 (placed)	Explosive	1 (single-use)	1.5
Bear trap	35 (burst)	100 (if triggered)	0 (placed)	Pierce/Blunt (trap)	5 resets	3.0

Spray / Flame

Weapon	Damage	Accuracy	Range	Type	Durability	Weight (kg)
Pepper spray	2 (+ blind/disable)	70	2m	Toxic	40 (mechanism; fuel tracked separately, ~30 uses)	0.3

Weapon	Damage	Accuracy	Range	Type	Durability	Weight (kg)
Aerosol flamethrower	30/sec (DoT stream)	75	4m	Burn	40 (nozzle mechanism; fuel tracked separately, see Ammo/Fuel table)	2.0

Armor Stats

Defense is flat physical damage reduction per hit (before Type-specific resistances). **Weight** affects stamina/mobility, not a hard number here – Light/Medium/Heavy. **Durability** degrades with hits taken, same jank-multiplier logic as weapons.

Armor	Defense	Weight	Durability	Coverage	Special Resistance
Padded jacket (impact)	8	Light (2.0 kg)	40	Torso/Arms	Minor Slash resist (+10%)
Reinforced boots	5	Light (1.0 kg)	45	Feet	None
Scrap plate vest	15	Heavy (6.0 kg)	50	Torso	None
Faraday-lined coat	10	Medium (3.0 kg)	35	Torso	High EMP resist (+50%)
Riot patchwork jacket	20	Medium (4.0 kg)	60	Torso/Arms	Blunt resist (+15%)
Thermoptic-lite cloak	5	Light (1.5 kg)	20	Full body	Stealth: chance to avoid detection while stationary/moving slow
Subdermal plating	25	None (internal, 0 kg carry weight)	80	Torso (internal)	Always-on, no equip slot; requires Ripperdoc to repair/replace

Design Notes

- No two items share a full stat line, but a few sit close on purpose (e.g. Zip gun vs. Improvised slingshot both low-tier ranged, but one trades accuracy for damage and durability for the other) – that overlap is what makes early crafting choices feel like trade-offs instead of a strict upgrade path.
- **Durability vs. Jank interaction:** a Rigged/Janky/Deathtrap build of any weapon above keeps its listed Damage/Accuracy/Range base stats but multiplies effective Durability loss per the Jank System's degradation rates (2x for Janky, higher for Deathtrap) – so the numbers here are the ceiling, not a guarantee.
- Consider letting **Repair kits** (Tier 1: tape + scrap metal; Tier 2: adds coolant/lubricant) restore a flat amount of Durability rather than a full reset, so weapon upkeep stays part of the scavenging loop.

- Armor **Coverage** could later support hit-location logic (aimed shots bypass boots/gloves, headshots ignore torso armor) if your combat system tracks hit zones.

Ammo Crafting

Guns that fire conventional projectiles need actual rounds — this closes the loop so “found a gun” and “found bullets for it” are separate scavenging problems, same as everywhere else in the system.

Core Components

- **Used shell casing** — the reusable brass/shell body. Recovered from battlefields, shooting ranges, or picked up after any firefight (yours or someone else’s).
- **Gunpowder** — the propellant. Scavenged from fireworks, hunting/ammo stores, or mixed at the Chem Lab from chemical scraps (saltpeter + charcoal + sulfur substitutes).
- **Primer** — small percussion cap that ignites the powder. Tier 2, found in ammo kits, gun stores, hardware stores.
- **Slug/shot** — the actual projectile: scrap metal melted into a slug, or ball bearings for shotgun buckshot.

Recipes

Ammo Type	Ingredients	Output	Station	Used By
Pistol/Rifle Round	Used shell casing + Gunpowder + Primer + Scrap metal (slug)	1 round (batch of 5 recommended)	Workbench, T2	Zip gun, Military rifle
Shotgun Shell	Used shell casing + Gunpowder + Ball bearings (buckshot)	1 shell (batch of 4 recommended)	Workbench, T2	Sawn-off shotgun
Railgun capacitor charge	None — draws from the weapon’s own Fusion Cell	8 shots per full charge	Recharges at Tech Rig (see Fusion Cell Recharging)	Railgun coil pistol

Nail gun (nails), crossbow (bolts), and slingshot (ball bearings/rocks) already have their own simple, gunpowder-free ammo loops covered earlier — they don’t need this system.

Jank Applies to Ammo Too

A batch of ammo crafted with a substituted component (e.g., homemade gunpowder instead of scavenged, or a scrap-metal slug poured without a proper mold) carries its own small Jank Rating, separate from the weapon firing it. Janky ammo can misfire (dud round, no damage) or, at worst, cause the *weapon* to take extra durability damage on that shot — a good reason to keep at least a few “clean” rounds in reserve for when reliability actually matters.

Batch Crafting

Ammo should craft in batches (5 rounds / 4 shells per action) rather than one at a time — otherwise the crafting menu becomes tedious busywork. Recommend showing “craft time” as a single batch cost (see below), not per-round.

Carry Weight & Encumbrance

Base Carry Capacity

- **Base limit: 40 kg.** Represents what a person can carry at a sustainable pace without penalty.
- **Augmented limit:** cybernetic frame/spine reinforcements (Ripperdoc upgrades) can raise this — suggested +10 kg per augment tier, up to a reasonable cap (e.g., 70 kg fully augmented).
- **Backpack/rig upgrades:** a crafted or looted carry rig (Tier 1: scrap frame + cloth straps) can add a flat +10–15 kg without needing cyberware.

Encumbrance Bands

Load (% of capacity)	Effect
0–80%	No penalty
81–99%	Burdened — movement speed ~15%, no sprint
100–120%	Overloaded — movement speed ~35%, no sprint or dodge, stamina drains while moving
120%+	Cannot move until weight is dropped

Design Notes

- Weapon/armor weight is listed per-item in the tables above; material weights are listed per unit in the Material Tiers tables (per unit, per handful, or per bag/coil as noted — keep these as your smallest tradeable unit so stacking math stays simple).
- Ammo is cheap by weight (0.02–0.03 kg/round) on purpose — the bottleneck should be *components*, not carry space, so players don't feel punished for stockpiling bullets specifically.
- Consider a **stash/base storage** system with no weight limit (or a much higher one) so players have a reason to establish a home base and make return trips, rather than just carrying everything found forever.

Craft Time

Craft time scales with station and tier, not with every individual recipe — keeps the numbers predictable for players without needing a bespoke timer on all 40+ items.

Baseline by Station & Tier

Station / Tier	Craft Time
Scrap Bench (T1)	15–30 sec
Workbench (T1)	1–2 min
Workbench (T2)	3–6 min
Chem Lab (T1)	2–4 min

Station / Tier	Craft Time
Chem Lab (T2/T3)	5–15 min
Tech Rig (T2)	5–10 min
Tech Rig (T3)	10–20 min
Ripperdoc (T2/T3, cybernetics)	20–60 min (surgery — may require a "resting" downtime period rather than being watchable in real time)
Ammo batch (5 rounds / 4 shells)	1 min
Fusion cell recharge	10–15 min (or instant if you'd rather it just be a resource cost)

Notable Exceptions (complex assemblies that don't fit the curve)

Item	Craft Time	Why it's an outlier
Jury-rigged chainsaw	8 min	More assembly than a typical T2 Workbench item
Shaped charge	6 min	Careful, deliberate assembly despite being "just" T2
Smart mine	15 min	AI-core integration on top of standard mine assembly
Railgun coil pistol	20 min	High end of T3 Tech Rig complexity
Subdermal plating	45 min	Full surgical implant procedure
Repair/tune pass (any item)	3–10 min, scaling with tier	Field maintenance, not a full rebuild

Design Notes

- Jankier builds should NOT be faster — if anything, a slight time penalty (+10–20%) for freehanding without a schematic reinforces that jank comes from working carelessly, not from working fast.
- Interruptible crafting (an enemy encounter mid-build) fits naturally at Workbench/Scrap Bench speeds; Ripperdoc and long Tech Rig builds are better gated to safe locations only, given their length.

Skill Progression — Crafting Skill Trees

Rather than one generic tree, crafting skill splits into **five branches**, one per specialization, each fed by its own XP pool earned through that branch's actions. This lets a player's build actually diverge — a dedicated Chemist plays differently than a Ripperdoc specialist, even though both are "crafters."

XP Sources (per branch)

Branch	Station(s)	XP Sources
Mechanical	Scrap Bench + Workbench	Crafting/repairing weapons, armor, ammo
Chemistry	Chem Lab	Crafting meds, drugs, explosives, gas grenades
Engineering	Tech Rig	Crafting electronics, drones, hacking tools; recharging fusion cells
Cybernetics	Ripperdoc	Installing/removing implants
Scavenging	Field (no station)	Successful searches, first-time location clears, schematic discovery

Universal XP rule across all branches: T1 craft = 5 XP, T2 = 15 XP, T3 = 30 XP; tuning a Janky item down a band = +10 XP in that item's branch; surviving a catastrophic failure without dying = +5 XP.

1. Mechanical (Scrap Bench + Workbench)

Level	Perk	Effect
1	Scrapper's Instinct	-5 Jank on all Scrap Bench builds
2	Steady Hands	-10 Jank specifically from material substitutions
3	Efficient Workspace	-15% craft time at Scrap Bench/Workbench
4	Field Engineer	Halves the "no schematic" Jank penalty (+10 → +5)
5	Stabilizer Specialist	Stabilizer Components cost 50% fewer materials; their Jank reduction improves from -15 to -20
6	Steady Grip	-50% Handling Risk (self-injury chance) on all weapons
7 (capstone)	Master Fabricator	Flat -20 Jank on all builds regardless of substitutions; repair/tune passes can drop an item two Jank bands instead of one

2. Chemistry (Chem Lab)

Level	Perk	Effect
1	Home Cook	-10 Jank on all Chem Lab T1 builds
2	Steady Measure	Explosive compounds get +10% blast reliability (lower dud chance on pipe/nail bombs)
3	Potent Mix	Drugs and meds are 15% more effective (potency or duration) with no material change
4	Ventilation Sense	Halves self-inflicted inhalation/backfire risk from gas and fire weapons (choke-gas bottle, aerosol flamethrower)

Level	Perk	Effect
5 (capstone)	Master Chemist	Once per in-game day, substitute one tier down on any Chem Lab recipe with zero Jank penalty

3. Engineering (Tech Rig)

Level	Perk	Effect
1	Solder Sense	~10 Jank on Tech Rig T2 builds
2	Signal Clarity	Electronics (jammers, scopes, comms, drones) get +10% effective range/accuracy
3	Overclocker	~20% fusion cell recharge time
4	AI Whisperer	Halves Jank specifically on AI-core-based builds (drones, autopilot, smart mines)
5 (capstone)	Master Engineer	Flat ~20 Jank on Tech Rig T3 builds; crafts one free Stabilizer Component per in-game day at no material cost

4. Cybernetics (Ripperdoc)

Level	Perk	Effect
1	Steady Scalpel	~15% surgery time
2	Sterile Technique	~15% implant rejection/malfunction (cyberware Jank) chance
3	Neural Compatibility	~10 flat Jank on all Ripperdoc builds
4	Quick Recovery	Halves post-surgery "resting" downtime
5 (capstone)	Master Ripperdoc	Can install Tier 3 implants using only a Tier 2 prerequisite material once unlocked — a genuine shortcut, not just a jank reduction

5. Scavenging (Field)

Level	Perk	Effect
1	Sharp Eyes	+10% find chance at all scavenge locations
2	Efficient Search	Reduced search time per location
3	Pack Mule	+10 kg carry capacity
4	Appraiser's Eye	See an item's Jank rating before looting it (off a corpse, a container, etc.)

Level	Perk	Effect
5 (capstone)	Master Scavenger	Locations repopulate loot faster; any search has a chance to yield a bonus Tier 2/3 material regardless of that location's normal weighting

Design Notes

- This is additive investment, not a single build path — a "gutter tech" character might dump points into Mechanical + Scavenging, while a support-focused player goes Chemistry + Cybernetics. Nothing stops a generalist, it's just slower to feel the payoff.
- **Appraiser's Eye (Scavenging 4)** and **Master Chemist's** one-per-day substitution both give a skilled player *player agency over risk* rather than just flatly better numbers — a nice way to make late-game crafting feel skill-expressive instead of just bigger stats.
- **Ventilation Sense** and **Steady Grip** both target the Handling Risk table directly — worth flagging in tooltips ("reduces self-injury chance") so players connect the dots between the skill tree and the mechanic it's fixing.
- None of these branches unlock new recipes — that stays behind schematics, keeping "what you can build" and "how well you build it" as separate, legible systems.

Unified Tree Diagram & Hybrid Nodes

The five branches aren't fully isolated — a handful of **hybrid nodes** sit at intersections and require investment in two branches at once, turning five parallel lines into an actual tree with convergence points. These don't reduce Jank further; they unlock small cross-discipline conveniences that wouldn't make sense in a single branch.

```
graph TD
    subgraph MECH[Mechanical]
        M1[Scrapper's Instinct] --> M2[Steady Hands] --> M3[Efficient Workspace] -->
        M4[Field Engineer] --> M5[Stabilizer Specialist] --> M6[Steady Grip] --> M7((Master Fabricator))
    end

    subgraph CHEM[Chemistry]
        C1[Home Cook] --> C2[Steady Measure] --> C3[Potent Mix] --> C4[Ventilation Sense] -->
        C5((Master Chemist))
    end

    subgraph ENG[Engineering]
        E1[Solder Sense] --> E2[Signal Clarity] --> E3[Overclocker] --> E4[AI Whisperer] -->
        E5((Master Engineer))
    end

    subgraph CYB[Cybernetics]
        Y1[Steady Scalpel] --> Y2[Sterile Technique] --> Y3[Neural Compatibility] -->
        Y4[Quick Recovery] --> Y5((Master Ripperdoc))
    end

    subgraph SCAV[Scavenging]
        S1[Sharp Eyes] --> S2[Efficient Search] --> S3[Pack Mule] --> S4[Appraiser's Eye] -->
        S5((Master Scavenger))
    end

    M5 -. -> H1{{Weaponsmith}}
    E3 -. -> H1
    C3 -. -> H2{{Field Medic}}
    Y2 -. -> H2
    M2 -. -> H3{{Trap Master}}
```

S3 -. -> H3
E4 -. -> H4{{Ghost Rig}}
S4 -. -> H4

Hybrid Node	Requires	Effect
Weaponsmith	Mechanical 5 + Engineering 3	Tech Rig-tier weapons (railgun coil pistol, smart-tracking dart launcher) can be built at the Workbench instead, with an extra -10 Jank
Field Medic	Chemistry 3 + Cybernetics 2	Nanite healing gel can substitute biotech tissue with standard chemical scraps at no Jank penalty; healing items apply 20% faster
Trap Master	Mechanical 2 + Scavenging 3	Bear traps and mines gain +25% placement stealth (harder for enemies to spot) and reset/rearm using fewer materials
Ghost Rig	Engineering 4 + Scavenging 4	Thermoptic-lite cloak crafts with -15 flat Jank, and its detection-avoidance bonus applies while moving slowly, not just stationary

Hybrid nodes are deliberately few — the goal is a couple of memorable "oh, I can combine these" moments late-game, not a dense web that forces every player down the same optimal path.

Economy — Barter & Coin Value

There's no hard currency — everyone trades goods for goods. **Coin** is an abstract accounting unit (not a physical object) used to price every item consistently so trades can be resolved as "your scrap metal + batteries vs. their steel plating," rather than needing a barter-value chart for every possible pairing. Think of it as what both sides silently convert to in their heads before agreeing a trade feels fair.

How Pricing Works

Every item has a **Base Value (BV)** — its "fair trade" worth. From that:

- **Sell Range** (what a merchant pays you) = **40–70% of BV**, depending on that merchant's interest in the item.
- **Buy Range** (what a merchant charges you) = **120–180% of BV**, depending on that merchant's markup.

The spread exists because no single merchant sits at the extremes — a Junk Dealer sells scrap cheap but won't pay much for your fusion cell; a Black Market Fixer pays well for that fusion cell but barely looks at your scrap metal. The per-item tables below show the full possible range across all merchant types; the Merchant Types table shows where each one actually sits.

Merchant Types

Merchant	Specialty (best rates)	Buy Markup	Sell Rate	Honesty	Haggle Room	Shortchange Chance
Junk Dealer	Tier 1 materials	130% (cheap)	55% (T1) / 30% (T2–3, not interested)	High	10%	0%
Corner Store Vendor	Consumables, chem	120% (cheap)	50% flat	High	5% (fixed)	0%

Merchant	Specialty (best rates)	Buy Markup	Sell Rate	Honesty	Haggle Room	Shortchange Chance
	ingredients, ammo parts				prices)	
Gang Quartermaster	Weapons, ammo (doesn't care about Jank)	140%	60% (weapons/ammo) / 25% (other)	Low	25%	25%
Ripperdoc	Cybernetic components, implants	170% (expensive)	70% (cyber) / 15% (other)	Medium– High	15%	5%
Black Market Fixer	Tier 3 materials, rare tech, schematics	160%	65% (T2–3) / 20% (T1, not interested)	Medium	20%	10%
Traveling Caravan Trader	Nothing specific – broad, decent rates	135%	55% flat across tiers	High	20%	0%

Honesty & Hagging

- **Haggle Room** is the max % a successful negotiation can move the listed price, in the player's favor. A Barter/Persuasion skill check (or standing reputation with that merchant/faction) determines success; failure leaves the price as listed rather than making it worse.
- **Shortchange Chance** is a hidden risk on dishonest merchants: after a deal is agreed, there's a chance the actual payout comes in 10–30% under what was agreed (fewer materials handed over, a worse-condition item substituted). **Appraiser's Eye** (Scavenging skill, Level 4) lets a player catch this before finalizing; otherwise it's only obvious after the fact, which is the point – it's a trust mechanic, not just a math tax.
- High-Honesty merchants (Junk Dealer, Corner Store, Traveling Trader) never shortchange, but also have small Haggle Room – what you see is close to what you get.
- Reputation with a merchant or their faction should nudge these numbers over time (better Sell Rate and lower Shortchange Chance as trust builds), giving players a reason to keep trading with the same fence rather than shopping around every time.

Item Prices – Materials (Tier 1)

Item	Base Value	Sell Range	Buy Range
Scrap metal	1	1	1–2
Copper wire	1	1	1–2
Plastic scrap	1	1	1–2
Cloth/fabric	1	1	1–2
Glass shards	1	1	1–2
Rubber	1	1	1–2

Item	Base Value	Sell Range	Buy Range
Wood	1	1	1–2
Duct tape/adhesive	1	1	1–2
Chemical scraps	2	1	2–4
Bone/leather	2	1	2–4
Spring/coil	2	1	2–4
Razor blades	2	1	2–4
Razorwire (coil)	3	1–2	4–5
Lighter	2	1	2–4
Aerosol spray	2	1	2–4
Nails (handful)	1	1	1–2
Fuel canister	4	2–3	5–7
Ball bearings (bag)	2	1	2–4
Fishing line/cord	1	1	1–2
Bleach	2	1	2–4
Matches	1	1	1–2
Gunpowder (unit)	3	1–2	4–5
Used shell casing	1	1	1–2

Item Prices — Tools

Item	Base Value	Sell Range	Buy Range
Wire cutters	6	2–4	7–11
Soldering iron	8	3–6	10–14
Screwdriver/wrench set	5	2–4	6–9

Item Prices — Materials (Tier 2)

Item	Base Value	Sell Range	Buy Range
Steel plating	10	4–7	12–18
Circuit boards	8	3–6	10–14

Item	Base Value	Sell Range	Buy Range
Batteries	10	4–7	12–18
Small motors/servos	12	5–8	14–22
Optics/lenses	12	5–8	14–22
Fiber optic cable	9	4–6	11–16
Coolant/lubricant	7	3–5	8–13
Explosive compounds	18	7–13	22–32
Reinforced polymer	14	6–10	17–25
Basic chips/processors	8	3–6	10–14
Capacitors	9	4–6	11–16
Primer	6	2–4	7–11

Item Prices – Materials (Tier 3)

Item	Base Value	Sell Range	Buy Range
AI cores/neural chips	80	32–56	96–144
Cybernetic components	70	28–49	84–126
Fusion/energy cells	60	24–42	72–108
Nanite paste	55	22–39	66–99
Smart-alloy	65	26–46	78–117
Quantum/encrypted chips	75	30–53	90–135
Biotech tissue samples	50	20–35	60–90
Graphene sheeting	45	18–32	54–81
Holo-emitters	40	16–28	48–72

Item Prices – Weapons (Melee)

Weapon	Base Value	Sell Range	Buy Range
Pipe shiv	8	3–6	10–14
Broken-bottle knife	6	2–4	7–11
Nail bat	12	5–8	14–22

Weapon	Base Value	Sell Range	Buy Range
Razorclub	14	6–10	17–25
Fishing-line garrote	9	4–6	11–16
Razor whip	16	6–11	19–29
Stun baton	25	10–18	30–45
Taser gauntlet	22	9–15	26–40
Jury-rigged chainsaw	45	18–32	54–81
Monowire garrote	90	36–63	108–162

Item Prices — Weapons (Ranged)

Weapon	Base Value	Sell Range	Buy Range
Improvised slingshot	7	3–5	8–13
Zip gun	10	4–7	12–18
Improvised crossbow	20	8–14	24–36
Jury-rigged nail gun	18	7–13	22–32
Sawn-off shotgun	35	14–25	42–63
Smart-tracking dart launcher	70	28–49	84–126
Railgun coil pistol	140	56–98	168–252
Military rifle (<i>loot-only, rare</i>)	160	64–112	192–288

Item Prices — Thrown, Placed & Explosive

Weapon	Base Value	Sell Range	Buy Range
Choke-gas bottle	6	2–4	7–11
Molotov cocktail	8	3–6	10–14
EMP grenade	20	8–14	24–36
Pipe bomb	15	6–11	18–27
Nail bomb	18	7–13	22–32
Proximity mine	22	9–15	26–40
Smart mine	45	18–32	54–81

Weapon	Base Value	Sell Range	Buy Range
Shaped charge	35	14–25	42–63
Bear trap	12	5–8	14–22

Item Prices — Spray/Flame & Armor

Item	Base Value	Sell Range	Buy Range
Pepper spray	5	2–4	6–9
Aerosol flamethrower (device)	20	8–14	24–36
Padded jacket	10	4–7	12–18
Reinforced boots	6	2–4	7–11
Scrap plate vest	18	7–13	22–32
Faraday-lined coat	22	9–15	26–40
Riot patchwork jacket	28	11–20	34–50
Thermoptic-lite cloak	60	24–42	72–108
Subdermal plating (<i>pre-install kit only — can't resell once implanted</i>)	120	48–84	144–216

Item Prices — Ammo (sold/bought per batch)

Item	Batch Size	Base Value	Sell Range	Buy Range
Pistol/Rifle Round	5	5	2–4	6–9
Shotgun Shell	4	6	2–4	7–11
Fusion cell recharge (service, no owned power source)	—	—	—	5–10 flat fee

Design Notes

- **Jank affects sale price** — a Rigged/Janky/Deathtrap weapon or piece of gear should sell for proportionally less (suggest scaling the Sell Range down by the same % the Jank band cuts stats: Rigged ~10%, Janky ~25%, Deathtrap ~45%). Most merchants will notice visible jank on sight; only a dishonest one might try to buy it cheap and resell it as clean.
- **Faction/reputation discounts** stack on top of the base Merchant table rather than replacing it — a trusted Gang Quartermaster might effectively act like a higher-Honesty merchant for that player specifically, without changing the archetype for everyone else.
- **The Black Market Fixer being reputation-gated** gives Tier 3 material trading a soft floor early game — players can find T3 mats in the world, but converting a stockpile into coin/other-goods efficiently is a

mid-to-late-game unlock, which paces the economy alongside the crafting tiers.

- Consider a soft **daily haggle limit** per merchant (they'll only negotiate down so far, so many times, before "final offer, take it or leave it") so haggling stays a meaningful choice rather than something to spam every transaction.

Commercial Locations

Merchants don't need dedicated maps of their own — several either reuse a location already on the Scavenge Locations list (guarded while the merchant's active there) or introduce a new stop along the map. A few locations even have **two states**: a functioning commercial version and a fallen/raided version that becomes lootable once the merchant's gone, which is a nice piece of environmental storytelling for free.

Commercial Location	Merchant(s) Present	Also Scavengable?	Overlaps With (Scavenge Table)	Risk
Mall (partially operational)	Corner Store Vendor (multiple stalls)	Yes — collapsed/closed wings scavenge like an Electronics store (circuit boards, chips, optics)	New location	Low in the operating wing (patrolled); Medium in the collapsed wing (squatters, structural risk)
Junkyard Fence Shack	Junk Dealer	Yes — this <i>is</i> the Junkyard	Junkyard	Low–Medium (same as Junkyard entry)
Standalone Corner Store / Bodega	Corner Store Vendor	Lightly — a few consumables/chem items behind the counter if the vendor's hostile or absent	New location (small settlements, shack clusters)	Low
Gang Den / Clubhouse	Gang Quartermaster	Yes, if you fight through instead of trade	Gang den/clubhouse (raided)	High if hostile; Low if trading peacefully
Chop Shop	Junk Dealer or Gang Quartermaster (depends on which faction currently controls it)	Yes — existing Chop Shop loot table	Chop shop	Medium
Ripperdoc Clinic (active)	Ripperdoc	No — protected business, guarded while operational	<i>Becomes</i> Ripperdoc back-room (raided) if the clinic falls	Low (bouncers deter casual theft)
Black Market Den	Black Market Fixer	No — secure, reputation-gated access, not a combat location	New location	N/A (access-gated, not fought into)

Commercial Location	Merchant(s) Present	Also Scavengable?	Overlaps With (Scavenge Table)	Risk
Caravan Stop / Trade Road waypoint	Traveling Caravan Trader (periodic — rotates through on a schedule, not always present)	Partial — a wrecked prior caravan nearby is scavengable if the trader was previously raided by someone else	New location	Low while the trader's present; Medium when it's just an abandoned wreck

Design Notes

- **The Ripperdoc pairing is the clearest "two states" example:** the active Clinic is guarded and off-limits to looting, but if it gets destroyed or abandoned over the course of the story, it becomes the already-defined "Ripperdoc back-room (raided)" scavenge location — same place, before and after. Consider doing the same with a Corp Office (active, has an Operative/security presence and no looting) vs. a Corp Office (abandoned, on the original Scavenge Locations table) if you want another instance of this.
- **Chop Shop's dual ownership** is a good hook for faction conflict — controlling it could be a small territory objective, and which merchant runs it afterward changes what the location is good for (Junk Dealer = cheap parts volume, Gang Quartermaster = better weapon resale).
- **Black Market Fixer being non-scavengable is deliberate** — it's the one node in the economy that's pure trust/reputation, no combat shortcut, keeping the "convert big stockpiles of rare material" bottleneck meaningful even for a very combat-capable player.
- If the game supports it, letting player choices (which faction wins a territory fight, whether a merchant is killed vs. spared) move merchants between these locations, rather than locking them in place, would make the world feel like it's reacting to the player instead of just hosting fixed vendor stalls.

Repair Kits

Consumed on use to restore a flat amount of Durability to a weapon, armor piece, or (see Vehicles below) a vehicle. Repairing does **not** by itself reduce a Jank band — that still requires a separate Tune Pass (time at the matching station, optionally with a Stabilizer Component) as described in the Jank System, except where noted below.

Repair Kit	Ingredients	Station	Craft Time	Restores	Base Value	Sell Range	Buy Range
Basic Repair Kit	Duct tape + Scrap metal	Scrap Bench, T1	1 min	+15 Durability (any item)	6	2–4	7–11
Field Repair Kit	Scrap metal + Coolant/lubricant + Reinforced polymer	Workbench, T2	3 min	+30 Durability (any item)	15	6–11	18–27
Tech Repair Kit	Circuit board + Coolant/lubricant + Basic chip	Tech Rig, T2	5 min	+30 Durability, specifically to electronics/tech-tier items	20	8–14	24–36

Repair Kit	Ingredients	Station	Craft Time	Restores	Base Value	Sell Range	Buy Range
Nanite Repair Compound	Nanite paste + Smart-alloy	Tech Rig or Ripperdoc, T3	10 min	Full Durability restore and drops the item one Jank band	50	20–35	60–90
Vehicle Repair Kit	Steel plating + Coolant/lubricant + Scrap metal	Workbench, T2 (Garage base module halves material cost)	5 min	+50 Durability (vehicles only)	25	10–18	30–45

Design Notes

- The Nanite Repair Compound is the one premium option that touches Jank directly – a deliberate “spend big to fix it properly” release valve for a Deathtrap-tier item someone’s grown attached to.
- Repair kits should be a natural **Corner Store / Junk Dealer** stock item (Basic/Field tiers) and a **Ripperdoc/Black Market** stock item (Nanite Compound), giving every merchant type a reason to carry at least one SKU relevant to their specialty.

Base Building

Several systems already assume a home base exists – Stabilizer Components, the Tech Rig’s fusion cell recharging, and stash storage all need somewhere to live. Here’s the shape of it.

Claiming a Base

A base is a location the player clears and holds – several entries from the Scavenge Locations table double as viable base sites once their occupants are dealt with, each with a natural capacity/risk trade-off:

Base Site	Capacity	Base Risk	Notes
Shack / squatter camp	Small (2–3 modules)	Low	Cheap, fast first base; not much room to grow
Abandoned house	Small–Medium (3–4 modules)	Low	Good early base, close to city locations
Parking garage	Medium	Low–Medium	Best natural fit for a Garage module (vehicles)
Abandoned factory	Large (6+ modules)	Medium (must clear occupants)	The long-term base – room for everything
Chop shop (via faction takeover)	Medium	Medium	Doubles as a commercial location – claiming it is a territory objective, not just a scavenger run

Base Modules

Module	Function	Build Materials	Build Time	Base Tier
Scrap Bench	On-site Scrap Bench crafting	Wood + Scrap metal	10 min	1
Storage Rack	+50 kg stash capacity (stackable)	Scrap metal + Wood	15 min	1
Bed / Rest Area	Enables downtime rest (surgery recovery, time-skip)	Cloth + Wood	10 min	1
Perimeter Fence	Reduces random-encounter chance at base (uses the existing Barbed Scrap Fence recipe)	Scrap metal + Wood	15 min	1
Workbench	On-site Workbench crafting	Steel plating + Wood + Tools	30 min	2
Power Generator	Provides a Power Pool; consumes 1 fuel canister/in-game day	Small motor + Scrap metal + Coolant	20 min	2
Motion Alarm	Early warning of approaching threats (existing recipe)	Circuit board + Battery + Basic chip	20 min	2
Garage	Vehicle storage/repair; halves Vehicle Repair Kit material cost	Steel plating + Scrap metal + Tools	40 min	2
Chem Lab	On-site Chem Lab crafting	Glass shards + Chemical scraps + Coolant	30 min	2
Tech Rig	On-site Tech Rig crafting + fusion cell recharging	Circuit boards + Capacitors + Basic chips	45 min	3 (needs Power Online)
Solar Panel	Smaller, free Power Pool; only recharges by day	Optics/lenses + Circuit board + Reinforced polymer	30 min	3
Auto-Turret	Active base defense (existing recipe)	Steel plating + Small motor + AI core + Fusion cell	60 min	3 (needs Power Online)
Ripperdoc Chair	On-site cybernetics installation	Cybernetic components + Steel plating + Nanite paste	60 min	4 (needs Power Online)

Power System

Generators/solar panels feed a **Power Pool** (an abstracted daily resource, not literal watts). Rough draw per action:

- Fusion cell recharge: ~20 units
- A Tier 3 Tech Rig craft: ~10 units
- Auto-turret active during an attack: ~5 units

- Passive lighting/alarms: ~5 units/day

A basic Generator refills the pool once per day at the cost of 1 fuel canister; a Solar Panel refills a smaller pool for free but only while it's daylight. Modules that "need Power Online" (Tech Rig T3 crafts, Auto-Turret, Ripperdoc Chair) still technically function without power — they just carry a flat **+20 Jank penalty** to represent doing precision work by hand/flashlight instead of properly powered.

Base Tiers

Tier	Name	Unlocks
1	Camp	Scrap Bench, Storage Rack, Bed, Perimeter Fence
2	Workshop	+ Workbench, Generator, Motion Alarm, Garage
3	Fortified Base	+ Tech Rig, Chem Lab, Solar Panel, Auto-Turret
4	Stronghold	+ Ripperdoc Chair, upgraded (Fusion) Generator

Base Defense & Raids

A base can be targeted by a random raid event, weighted by **Territory Heat** (how much the player has antagonized nearby factions — ties into the Reputation system flagged as still-needed) and the site's Base Risk rating. Defenses mitigate rather than prevent:

- **Motion Alarm** — forewarns the player, allowing prep or evacuation.
- **Perimeter Fence** — slows attackers, buying time.
- **Auto-Turret** — fights automatically, but can run out of Power Pool or ammo mid-fight.

A lost defense results in **stolen stash items** (weighted toward light/valuable goods) and **damaged modules**, which drop to a non-functional state until repaired — the natural use case for the Repair Kits above, and a good reason those kits should be a base-stocking priority, not just a field item.

Design Notes

- The Chop Shop base option is the one deliberate overlap with the Commercial Locations system — claiming it as a base and running it as a merchant location aren't mutually exclusive, which gives factional territory control real mechanical weight.
- Consider letting an unpowered Tech Rig/Ripperdoc still be usable in early game (with the Jank penalty) rather than hard-gating them behind a Generator build — keeps a new base from feeling like dead weight for the first few hours.

Vehicles

The mods already defined (Reinforced bumper, Jury-rigged engine patch, Overclocked engine chip, Autopilot module, Nitro injector) need a base vehicle to attach to.

Base Vehicle Types

Vehicle	Speed	Handling	Durability	Fuel Type & Capacity	Cargo Capacity	Weight Class	Found At
Scrap Motorcycle	70	High	80	Gasoline, 1.5 canisters	+10 kg	Light	Junkyard, Back alley, Chop shop
Beater Sedan	40	Medium	150	Gasoline, 3 canisters	+30 kg	Medium	Parking garage, Chop shop
Cargo Van/Pickup	30	Low	220	Gasoline, 4 canisters	+80 kg	Heavy	Abandoned factory, Chop shop
Corp Interceptor <i>(loot-only, extremely rare)</i>	90	High	180	Fusion Cell (1 cell ≈ full tank)	+20 kg	Medium	Military checkpoint, server farm, drone/vehicle wreckage site

Vehicle Jank

Any vehicle found in the world (rather than fully rebuilt by the player) starts at **Rigged or Janky**, same bands and effects as weapons – reduced Speed/Handling/Durability, and a per-trip malfunction chance (stall, engine fire, blowout) instead of per-use. A Vehicle Repair Kit plus a Mechanical Tune Pass brings it down toward Stable the same way a weapon would be tuned; the Garage base module speeds this up.

Mod Effects (tying back to the earlier crafting list)

Mod	Effect
Reinforced bumper	+20 flat Durability
Jury-rigged engine patch	+15 Durability, -10 Jank if applied to a freshly-found vehicle
Overclocked engine chip	+15% Speed, but +10 Jank (overheat/stall risk)
Autopilot module	Enables sending the vehicle on an unmanned supply run to a known location, resolved over abstracted time (requires the AI core already in its recipe)
Nitro injector	+30% Speed for a short burst; consumes ~0.5 fuel canister per use

Vehicle Pricing

Vehicle	Base Value	Sell Range	Buy Range
Scrap Motorcycle	60	24–42	72–108

Vehicle	Base Value	Sell Range	Buy Range
Beater Sedan	90	36–63	108–162
Cargo Van/Pickup	120	48–84	144–216
Corp Interceptor	300	120–210	360–540 (rarely actually for sale — mostly a "if you find it, it's yours" asset)

Design Notes

- Cargo Capacity is a direct extension of the Carry Weight system — a run to the Abandoned Factory in a Cargo Van turns a multi-trip slog into one haul, which is a good reason to want one beyond raw speed.
- The Corp Interceptor running on Fusion Cells instead of gasoline ties it into the same rechargeable-power loop as high-tier weapons, so owning one meaningfully raises the value of investing in a base Generator/Solar setup.
- Fuel/canister consumption is deliberately left approximate (a canister ≈ a day's use or ~50 km, whichever your game's scale favors) since the right number depends on how big your map actually is.

Factions & Reputation

Referenced repeatedly by now — merchant discounts, Black Market access, Territory Heat for base raids, Chop Shop control — so here's the actual system underneath it.

The Factions

Faction	Sphere / Locations	Opposed To	Role
The Corp	Corp offices, server farms, military checkpoints	The Wreckers, Black Market	Security bots, Operatives, Soldiers; controls high-tech territory and the best Tier 3 loot
The Wreckers (raider gang)	Gang den, chop shop, back alleys	The Corp, Minutemen	Raiders, Burners; controls scrap-tier territory and the weapons/ammo trade
The Minutemen (local militia)	Shacks, small settlements	The Wreckers	Defends civilian territory; source of Minuteman encounters and schematics
Scrapper's Guild (loose fence network)	Junkyard, city dump	None (opportunistic, trades with everyone)	Junk Dealers, Scavengers; neutral ground, the "everyone needs them" faction
The Fixers (black market network)	Black Market Den, ripperdoc back-channels	The Corp	Gatekeeps the Black Market Fixer and rare cyberware; reputation-gated by design

Reputation Scale

Each faction tracks a separate score from **-100 to +100**.

Range	Band	Effect
-100 to -60	Hated	Shoot on sight in their territory; possible bounty/hunted status
-59 to -20	Hostile	Aggressive on sight, but won't actively hunt you elsewhere
-19 to 19	Neutral	Standard merchant rates, no special treatment
20 to 59	Friendly	Better Sell Rate / lower Buy Markup with that faction's merchants (~+10% each direction); minor quests open up
60 to 100	Allied	Major discounts, faction-specific gear/schematics unlock, may assist in combat encounters in their territory

How Reputation Moves

- **Gains:** completing a faction's quests, defending their territory during a raid, selling them loot they specifically want, paying a one-time "in" (bribe/tribute).
- **Losses:** killing faction members, raiding their commercial/base locations, siding with an opposed faction in a conflict, being caught stealing.
- **Opposed factions move in opposite directions automatically** — raising standing with the Corp by, say, clearing Wreckers out of a checkpoint should dock Wrecker reputation without a separate action, so the player feels the tradeoff of picking a side rather than needing to grind two meters independently.

Interactions With Existing Systems

- **Merchant rates** — a merchant's actual Buy Markup/Sell Rate shifts based on the player's standing with that merchant's faction, layered on top of the base Merchant Types table rather than replacing it.
- **Black Market access** — requires Fixers reputation to reach at least Friendly before the Fixer will deal at all; this is the reputation gate already flagged on that merchant.
- **Territory Heat / base raids** — Heat against a given faction rises as reputation drops; Hostile or Hated standing with a faction whose territory is near the player's base meaningfully increases raid frequency from that faction specifically (a Wreckers-Hated player should expect Wreckers raids, not random Corp patrols).
- **World behavior** — Allied-band NPCs of that faction may wave the player through a checkpoint, warn them off traps, or simply not attack on sight even when otherwise "hostile" by location.

Design Notes

- Keep this to five factions rather than expanding further — enough to force real tradeoffs (Corp vs. Wreckers is the central one) without turning reputation management into its own spreadsheet.
- The Scrapper's Guild being permanently neutral is deliberate: it keeps the Junkyard/Dump loop (the game's early-game bread and butter) accessible no matter which side the player's picked, so a heavily Wrecker-aligned or heavily Corp-aligned character never gets locked out of basic scavenging.
- Consider a small passive reputation decay toward Neutral over time for factions the player hasn't interacted with recently, so old choices don't permanently lock in a world state if the player wants to pivot later.

This covers everything flagged as a gap. Let me know if you want any section revised, numbers rebalanced, or a new system entirely.

Combat Resolution

Ties Damage/Accuracy/Range/Type (Weapon Stats), Defense/Coverage (Armor Stats), and Jank/Handling Risk (crafting systems) into one resolution order, plus a hit-location system the existing Armor Coverage column was already implying.

Turn Order (per attack)

1. **Jank malfunction check** — roll the weapon's current Jank-band malfunction chance. If it triggers, resolve the Malfunction Outcome instead of a normal attack (jam, backfire, dud, etc.) and skip the rest of this sequence.
2. **Handling Risk check** — if the weapon carries one, roll it independently. A hit here damages the wielder regardless of whether the attack itself lands.
3. **To-Hit roll** — Accuracy vs. target Evasion (below). Miss ends the sequence; ammo/durability is still consumed per the weapon's normal use.
4. **Hit Location roll** — determines which Coverage slot (if any) absorbs the hit.
5. **Critical/Backstab check** — flat chance, or guaranteed on a successful stealth backstab.
6. **Damage calculation** — Weapon Damage × variance, reduced by Armor Defense at that location, then adjusted by the Type Interaction table.
7. **Status effect application** — if the weapon's Type inflicts one (Burn, Bleed, Stun, Blind, Disable).

To-Hit & Evasion

$\text{Hit Chance} = \text{Weapon/Enemy Accuracy} - \text{Target Evasion}$, clamped between 5% and 95% (nothing is ever a guaranteed hit or guaranteed miss).

Evasion by Speed (enemies) / **armor weight class** (player):

Source	Evasion
Fast (enemy Spd)	+20
Erratic (enemy Spd)	+15
Normal (enemy Spd)	+10
Slow (enemy Spd)	+0
Player in Light armor	+15
Player in Medium armor	+5
Player in Heavy armor	-5
Player unarmored	+10

Hit Locations

Location	Chance	Notes
Torso	40%	Covered by most armor (vest, jackets, coats)

Location	Chance	Notes
Arms	20%	Covered by jacket-type armor with "Torso/Arms" coverage
Legs	20%	Currently uncovered by any armor in this list – a clear gap if you want to add leg armor later
Feet	10%	Only Reinforced boots cover this
Head	10%	No armor in this list covers Head – headshots currently always bypass armor entirely. Deliberate high-risk/high-reward for aimed shots; add a helmet/mask item if you want players able to mitigate this

An uncovered location (or one outside the equipped armor's Coverage) takes full damage with **zero Defense reduction** – Coverage isn't just flavor text, it's the difference between "protected" and "not."

Damage Type Interactions

Type	vs. Armor Defense	Special
Blunt	Normal	+20% damage vs. Bot-type "light chassis" enemies
Slash	Normal	Small chance to inflict Bleed (DoT)
Stab/Pierce	Ignores 25% of target's Defense	—
Ballistic	Normal	–20% damage vs. Bot-type "heavy/plated" chassis unless noted armor-piercing
Explosive	Defense only half-effective	AoE radius; chance to Stagger (brief accuracy penalty on target's next action)
Burn	Ignores Defense entirely	Applies Burn DoT for 3 turns
Shock	Defense only half-effective	Chance to Stun (skip a turn); double effective damage vs. all Bot types
EMP	Minimal/no damage to Human/Animal	Severe damage + chance to fully Disable Bot-type enemies for several turns – the intended payoff for crafting EMP grenades/tasers at all
Toxic/Gas	Bypasses Defense entirely	Mostly inflicts Blind/Disable status rather than direct damage

Status Effects

Status	Effect	Typical Source
Bleed	Small DoT, several turns	Slash crits
Burn	Moderate DoT, 3 turns	Molotov, aerosol flamethrower
Stun	Skip next turn	Shock weapons, stun baton/taser

Status	Effect	Typical Source
Blind	~50% Accuracy, several turns	Choke-gas bottle, pepper spray
Disable (Bot only)	Fully incapacitated, several turns	EMP grenade, EMP-type effects
Tremor	~20% Accuracy, brief	Jank-malfunctioning cyberware (ties to the Jank System's cyberware malfunction outcome)

Critical Hits & Backstabs

- **Standard crit:** 5% flat chance on any hit, 1.5× damage.
- **Headshot:** rolled via the Hit Location table above – 2× damage, bypasses all current armor (see gap noted above).
- **Stealth backstab:** an attack landed while fully undetected is a guaranteed crit at 2.5× damage – this is the natural hook for the not-yet-built Stealth skill tree, which should be able to push detection avoidance and backstab multiplier further.

Worked Example

Player attacks a **Raider** (60 HP, 10 Armor/scrap plating, Torso/Arms coverage implied) with a **Sawn-off shotgun** (38 Damage / 50 Accuracy, currently Rigged-band Jank at ~90% stats):

1. Jank check: Rigged malfunction chance ~3% – rolls clean, no malfunction.
2. No Handling Risk on this weapon – skip.
3. To-Hit: 50×0.9 (jank stat reduction) = 45 effective Accuracy, Raider Evasion (Normal speed) = 10 → Hit Chance 35%. Roll succeeds.
4. Hit Location: rolls Torso (40% band).
5. No crit, no backstab.
6. Damage: 38×0.9 (jank) = ~34, Type Ballistic (normal vs. Defense) – Raider's 10 Armor = **24 damage dealt**.
7. No status effect (Ballistic doesn't inflict one by default).

Raider drops from 60 → 36 HP from that hit.

Design Notes

- The **Legs and Head coverage gaps are real** – right now no craftable armor protects either, which makes leg/headshots strictly better for an attacker with decent Accuracy. That's a fine intentional trade-off (encourages aimed-shot play), but add a pants/greaves item and a helmet if you'd rather close it.
- **EMP's Bot-Disable interaction is the payoff** promised back in the Enemies section – worth surfacing clearly in-game (a visible "DISABLED" state) so players learn the tactic rather than just noticing bots take more damage.
- Jank and Handling Risk checks happening *before* the to-hit roll means a badly-built weapon can fail a player even on an attack that would've hit clean – that's intentional (jank is a tax on reliability, not just damage) but worth confirming it reads clearly in UI (a jam should look different from a miss).

Non-Crafting Skills

Same shape as the crafting branches (level up by doing the thing, not a separate grind), covering everything the crafting/combat/economy systems assume a character can *do* rather than *build*.

Melee Combat

XP: dealing damage and getting kills with melee weapons.

Level	Perk	Effect
1	Iron Grip	+5% Accuracy with all melee weapons
2	Weight Behind It	+10% damage with Blunt-type weapons
3	Opportunist	+15% crit chance attacking a Stunned/Blinded/Disabled target
4	Close Quarters	+10 Evasion while a melee weapon is equipped
5 (capstone)	Executioner	Stealth backstab multiplier increases from 2.5x to 3.5x

Ranged Combat

XP: dealing damage and getting kills with ranged/thrown weapons.

Level	Perk	Effect
1	Steady Aim	+5% Accuracy with all ranged weapons
2	Quick Reload	~30% time to reload or swap ammo
3	Marksmanship	Unlocks an "Aimed Shot" action — costs extra time but lets you bias the Hit Location roll toward a chosen zone
4	Demolitions	+20% blast radius/damage with thrown and placed explosives
5 (capstone)	Deadeye	Ranged crit chance rises from 5% to 15%; headshots deal an extra +50% on top of their existing 2x

Stealth

XP: successful sneaking, undetected kills, successful lockpicking/pickpocketing.

Level	Perk	Effect
1	Quiet Steps	+15% chance to remain undetected while moving
2	Shadow's Grip	Backstab crit no longer requires being fully undetected — partial detection still grants a reduced 2x
3	Lockpick Mastery	~50% lockpick time, higher success chance
4	Fade Away	Breaking line of sight for a few seconds drops enemy alert state faster

Level	Perk	Effect
5 (capstone)	Ghost	While crouched/stationary, base detection chance is halved – stacks with the Thermoptic-lite cloak's stealth bonus

Netrunning / Hacking

XP: successful hacks, disabling bots/turrets remotely, bypassing terminals.

Level	Perk	Effect
1	Script Kiddie	+10% hack success on basic terminals/locks
2	Backdoor	Can remotely hack a Rogue Security/Kill bot for a brief Disable – a hacking-based alternative to the EMP grenade's Type interaction
3	Overclock Signal	~20% cooldown on hacking abilities (turrets, cameras, drones)
4	Ghost in the Machine	Successfully hacking a camera/alarm can retroactively reduce Heat gained from a botched stealth attempt
5 (capstone)	Root Access	Once per encounter, hijack a Rogue Kill Bot to fight on your side temporarily

Social / Barter

XP: successful haggling, persuasion checks, faction reputation gained through dialogue.

Level	Perk	Effect
1	Silver Tongue	+5% Haggle Room with all merchants
2	Reliable Face	Halves Shortchange Chance from dishonest merchants
3	Persuasive	Unlocks dialogue options to de-escalate a Hostile-band encounter without fighting
4	Networker	+20% reputation gained from faction quests
5 (capstone)	Silver-Tongued Devil	Once per day, reset a merchant's "final offer" limit for a second negotiation attempt

Survival / Athletics

XP: traveling distance, surviving encounters at low HP, resisting environmental hazards.

Level	Perk	Effect
1	Iron Stomach	Reduced damage/duration from Toxic status (landfill, gas weapons)
2	Second Wind	Small passive HP regen out of combat

Level	Perk	Effect
3	Sure-Footed	Halves the movement penalty from being Overloaded (Carry Weight system)
4	Adrenaline Rush	Below 25% HP, gain +10 Evasion and +10% Accuracy for a short duration
5 (capstone)	Survivor's Resolve	Once per day, automatically survive a killing blow at 1 HP (long cooldown)

Cyberware Proficiency

XP: using installed cyberware in combat/utility situations. Complements the Cybernetics *crafting* branch (which governs installing implants) by governing how well you *use* them.

Level	Perk	Effect
1	Neural Sync	+10% effectiveness from all installed cyberware bonuses
2	Steady Signal	~20% duration on Tremor status from cyberware malfunctions
3	Overclock Tolerance	Can run one implant beyond the normal slot limit at a minor Jank cost
4	Integrated Reflexes	Reflex Booster implant's Accuracy/Evasion bonus increased an extra 10%
5 (capstone)	Cybernetic Mastery	Immune to Tremor status; all installed implants function as if one Jank band cleaner than they actually are

Design Notes

- **Ranged Combat's Marksman/Deadeye perks are what actually close the Head/Legs armor gap** flagged in Combat Resolution — without investment, aimed shots are a coin-flip; with it, they become a real build direction rather than a pure exploit.
- **Netrunning is the intended "fourth path"** into fights that are otherwise combat/stealth/social — Root Access in particular gives it a genuine late-game power fantasy (turning a mini-boss kill bot against its own side) without touching the weapon/armor economy at all.
- **Cyberware Proficiency deliberately doesn't touch installation cost or Jank-at-build-time** (that's the Cybernetics crafting branch's job) — keeping "can you build/afford it" and "how good are you with it once installed" separate mirrors the crafting-vs-combat split used everywhere else in this doc.
- None of these trees grant new recipes or unlock new locations — consistent with the crafting trees, skill investment here is about doing the existing things better, not gating content behind a level number.

Cyberware & Humanity

The genre-standard cost for stacking implants — every cyberware install chips away at a **Humanity** stat (100 → 0), and running it too low has real consequences rather than just being flavor text.

Humanity Cost by Implant

Implant	Base Humanity Cost
Basic neural link (comms only)	-5
Grip-strength servo arm	-8
Reflex booster implant	-12
Optic HUD implant	-12
Subdermal armor weave	-15
Subdermal plating	-15

Jank modifies the cost — a rushed or badly-substituted install is more traumatic on the body/mind than a clean one, on top of the mechanical malfunction risk it already carries:

Install Jank Band	Humanity Cost Multiplier
Stable	×1.0
Rigged	×1.1
Janky	×1.25
Deathtrap	×1.5

A Deathtrap-quality subdermal plating install (-15 base) costs **-23 Humanity** instead of -15 — one more reason a Master Ripperdoc's clean-build guarantee matters beyond just avoiding malfunctions.

Humanity Bands & Effects

Range	Band	Effects
76-100	Stable (Human)	No effects
46-75	Strained	-5% Haggle Room (people find you a little off); occasional unsettled NPC dialogue flavor
16-45	Fraying	-15% Haggle Room; Persuasive/Social de-escalation perks half as effective; small chance per intense combat encounter of a forced-aggression episode (attack the nearest target, ally or not, for one turn)
0-15	Cyberpsychotic	Frequent forced-aggression episodes; the Minutemen and Corp treat you as Hated on sight regardless of actual faction standing; hitting 0 can trigger a rare Cyberpsycho Hunter encounter — an elite squad sent to put you down, the genre's classic "you went too far" boss-tier event

Restoring Humanity

- **Therapy session** — paid downtime at a Ripperdoc or dedicated counselor NPC; slow and expensive, restores a flat amount over time rather than instantly.

- **Neuro-Stabilizer** (new craftable drug — Nanite paste + chemical compound, Chem Lab, T3) — doesn't restore actual Humanity, just suppresses the current band's effects for a limited duration. A stopgap for a hunt or a big negotiation, not a fix.
- **Removing an implant** at the Ripperdoc Chair refunds 50% of that implant's Humanity cost, but the mechanical benefit and the materials used to build it are gone — genuinely walking something back, not a free undo.
- There's a **soft floor**: Humanity can't passively recover above (100 - total currently-installed cost) — you can manage the damage, but you can't fully forget you're carrying that much chrome while it's still in you.
- **Sterile Technique** (Cybernetics crafting, Level 2) reduces Humanity cost by the same 15% it already applies to malfunction chance — professional installation is easier on the psyche, not just the hardware.

Design Notes

- This is the natural tension for anyone leaning hard into the Cybernetics crafting branch or Cyberware Proficiency skill tree — more chrome is mechanically stronger but pulls the character toward a state where factions, merchants, and eventually the world itself start treating them differently.
- The Cyberpsycho Hunter event is best used sparingly — a genuine "point of no return" consequence works if it's rare and telegraphed (Fraying-band warnings, NPC dialogue) rather than something that can blindside a player who wasn't paying attention to the stat.
- Consider letting a few **narrative/companion reactions** hook into the Humanity bands directly — an ally commenting when the player crosses into Strained or Fraying does more world-building work than a bar on a menu screen ever will.

Day-Night & Patrol Cycles

The last open system — ties stealth, merchant availability, base raids, and a few enemy behaviors already half-defined (Minutemen "after dark," Solar Panel daylight-only) into one clock.

Cycle Structure

A full cycle = 24 in-game hours, split into four bands: **Dawn** (6–8), **Day** (8–18), **Dusk** (18–20), **Night** (20–6). Map that to whatever real-time pacing suits your game (a common approach: 1 real minute ≈ 1 in-game hour, so a full cycle is ~24 real minutes).

Effects by Time of Day

Band	Detection/Stealth	Notable Effects
Day	Baseline detection range	Corner Store Vendors and the Mall's operating wing are open; Solar Panels generate power; civilian NPC traffic (Street kids, Scavengers) is higher; Minutemen hold fire unless directly provoked
Dusk	Slightly reduced detection	Transition window — merchants begin closing, patrol density starts dropping at Corp/Military locations
Night	~20% detection range (stealth easier; stacks with Ghost/Thermoptic cloak)	Wild dog packs and Wireheads/Burners are more active; Corner Store and Mall are closed (only Gang Quartermaster, Black Market Fixer, and a present Traveling Trader operate); Minutemen treat any approach as suspicious rather than waiting for hostile intent; Base raid chance is weighted higher
Dawn	Baseline, transitioning	Patrols resume full staffing; Solar Panels begin recharging

Patrol Cycles at High-Security Locations

Corp offices, military checkpoints, and server farms run **guard shift changes** roughly every 6 in-game hours, each with a short (~10-minute) vulnerability window where coverage thins out – fewer active Operatives/Soldiers on the floor, security bot patrol routes gap. This window is the natural opening for a Stealth or Netrunning-focused approach: **Backdoor** (Netrunning skill) is a better bet during a shift change, when system logs see less active monitoring.

Location-Specific Notes

- **Junkyard, city dump, sewer** – no meaningful day/night difference; already low-oversight locations.
- **Library, Mall (operating wing), Electronics store** – effectively Day-only; closed or unstaffed at Night, which also means unguarded if you're there to scavenge rather than shop.
- **Corp office, server farm, military checkpoint** – Day means full patrol staffing but predictable routes; Night means thinner coverage but the shift-change window is the real opportunity, not just "it's dark."
- **Base** – Night raid weighting stacks with Territory Heat from the Faction system; a base with poor defenses and high Heat is at its most exposed overnight.

Design Notes

- This finally gives the **Minutemen's existing "holds fire... after dark" behavior** and the **Solar Panel's daylight-only recharge** an actual system to hang off, instead of being standalone rules.
- The shift-change window is a good place to reward planning – a player who's scoped a Corp office once during the day (learning the patrol pattern) should be able to come back at the right moment and have an easier time, rather than Night just being a flat "easy mode" toggle.
- Consider letting **Solar vs. Generator base power** create a genuine day/night tradeoff: a Solar-only base runs low on Power Pool by early Night, right when raid risk climbs – a Generator smooths that out at the cost of ongoing fuel consumption.

Player Attributes

Six core attributes, 1–10 scale, that sit underneath every system already built – they're what the crafting/combat/social/hacking numbers have been implicitly assuming a character *has* all along.

The Six

Attribute	Governs
Body	HP, carry capacity, melee damage
Reflexes	Evasion, ranged handling/reload speed
Tech	Crafting Jank reduction, hacking success
Cool	Resistance to Stun/Fear, Humanity decay resistance
Charisma	Haggle Room, persuasion, reputation gain rate
Perception	Scavenging find chance, aimed-shot/headshot bonus

Point Buy

30 points to distribute, minimum 1 and maximum 10 per attribute (an even 5-across spread is the baseline "no strong lean" build). Presets below spend the same 30 points differently rather than getting more.

Derived Formulas

These sit as *modifiers on top of* the baseline numbers already established elsewhere in this doc — they don't replace the Encumbrance, Evasion, or Jank tables, they adjust them per-character.

Formula	Effect
Max HP = $50 + (\text{Body} \times 10)$	Baseline HP pool
Carry Capacity = $40 + (\text{Body} - 5) \times 3$ kg	Adjusts the 40 kg base from the Carry Weight system
Evasion modifier = $(\text{Reflexes} - 5) \times 2$	Adds to the base Evasion table in Combat Resolution
Ranged Accuracy modifier = $(\text{Perception} - 5) \times 1\%$	Stacks with weapon Accuracy and skill perks
Melee damage modifier = $(\text{Body} - 5) \times 2\%$	Applied to melee Weapon Stats
Crafting Jank modifier = $(\text{Tech} - 5) \times -2$ flat Jank	Stacks with skill-tree Jank reduction
Hack success modifier = $(\text{Tech} - 5) \times 2\%$	Stacks with Netrunning skill perks
Stun/Fear resistance = $(\text{Cool} - 5) \times 3\%$ duration reduction	Reduces Stun/Blind/forced-aggression status duration
Humanity cost modifier = $(\text{Cool} - 5) \times -1\%$	Stacks with Sterile Technique
Haggle Room modifier = $(\text{Charisma} - 5) \times 1\%$	Stacks with the Economy's per-merchant Haggle Room
Reputation gain modifier = $(\text{Charisma} - 5) \times 2\%$	Stacks with Networker (Social skill)
Scavenging find-chance modifier = $(\text{Perception} - 5) \times 1\%$	Stacks with the Scavenge Locations find chance

Design Notes

- Every derived formula is centered on 5 — an average-everything character sees zero net modifier anywhere, so all the baseline numbers already in this doc remain the "default human" case. Attributes only push a character away from that baseline.
- Deliberately no attribute governs the crafting skill trees' Jank reduction *directly* replacing them — Tech adds a flat modifier alongside Mechanical/Chemistry/Engineering perks, so a high-Tech character with zero skill investment is still worse than a dedicated crafter with average Tech. Attributes set a character's lean; skills set their expertise.

Preset Character Types

Seven archetypes, each spending the same 30 attribute points differently and leaning into a different corner of the systems above. All are starting points, not locked classes — nothing stops a Gutter Tech from picking up Netrunning skills later.

Preset	Body	Reflexes	Tech	Cool	Charisma	Perception	Focus
Gutter Tech	7	5	8	4	3	3	Mechanical + Scavenging; embraces Jank rather than avoiding it
Corpo Defector	4	5	7	6	6	2	Engineering + Social; starts Friendly with the Corp, Hostile with the Wreckers
Street Medic	3	4	7	7	5	4	Chemistry + Cybernetics; support-focused, Field Medic hybrid node synergy
Enforcer	9	6	3	5	3	4	Melee Combat + Mechanical; front-line brawler
Netrunner	2	5	9	5	4	5	Netrunning + Engineering; fragile, fight-avoidant, hijacks bots instead
Fixer	4	5	3	5	9	4	Social + Scavenging; best prices in the game, worst in a straight fight
Ghost	3	8	3	5	3	8	Stealth + Ranged Combat; sniper/infiltrator, avoids fights it doesn't start

Starting Gear & Standing (suggested)

Preset	Starting Weapon/Armor	Starting Reputation Lean
Gutter Tech	Pipe shiv, Padded jacket	Neutral all factions, Friendly Scrapper's Guild
Corpo Defector	Railgun coil pistol (Rigged-band, "liberated"), Faraday-lined coat	Friendly Corp, Hostile Wreckers
Street Medic	Trauma stim x3, Padded jacket	Neutral all, slight Friendly Minutemen
Enforcer	Nail bat, Scrap plate vest	Neutral all, slight Hostile Corp (known troublemaker)

Preset	Starting Weapon/Armor	Starting Reputation Lean
Netrunner	Portable hacking deck, no armor	Neutral all, Hostile Corp (flagged in their systems)
Fixer	Zip gun, Padded jacket	Friendly Scrapper's Guild + Fixers
Ghost	Improvised crossbow, Thermoptic-lite cloak (Rigged-band)	Neutral all

Design Notes

- **Corpo Defector and Netrunner both start Hostile/flagged with the Corp** but for different reasons (defection vs. known intrusion) — worth differentiating in how the Corp actually reacts (a defector might be quietly hunted; a flagged netrunner might just get harder security everywhere) rather than treating "Hostile" as one uniform state.
- **Ghost and Enforcer are the clearest mechanical opposites** (Reflexes/Perception vs. Body, Stealth/Ranged vs. Melee/Mechanical) and worth using as the two ends of a build-diversity sanity check when balancing future content — if a location or fight only works for one of these two, it probably needs a second solution path.
- None of these presets lock out any skill tree or crafting branch — they're a starting lean (attributes + gear + rep) and nothing else, so respeccing playstyle over time through skill investment should always remain viable.